

Class 31: Inner Products

Many geometric ideas like length, angle, and distance can be defined in abstract vector spaces by an inner product of two vectors. The model for an inner product is the dot product defined in $\mathbf{R}^n$.

Definition: The inner product of vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbf{R}^n$ is given by

Properties of inner products:

Length of a vector:

Find a vector of length 1 in the direction of (3,4).

Key idea: Two vectors are orthogonal (perpendicular) if $\mathbf{u} \cdot \mathbf{v} = 0$.

Practice: Let $\vec{u} = \begin{bmatrix} 5 \\ 6 \\ 7 \end{bmatrix}$. Let $W = \text{span} \{ \vec{u} \}$
and Let $V = \{ \vec{x} \in \mathbb{R}^3 : \vec{u} \cdot \vec{x} = 0 \}$

Problem #28: Given that $\mathbf{y}$ is orthogonal to $\mathbf{u}$ and $\mathbf{v}$, prove that $\mathbf{y}$ is orthogonal to the $\text{span}\{\mathbf{u}, \mathbf{v}\}$.

Definition: Let W be a subspace of $\mathbf{R}$. The set of all vectors in $\mathbf{R}$ that are orthogonal to every vector in W is called the orthogonal complement of W .

Example: Let W be the row space of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 4 & 7 \end{bmatrix}$
What is $W^\perp$?

Important connections

Theorem: If W is a subspace of $\mathbf{R}^n$, then $W^\perp$ is also a subspace of $\mathbf{R}^n$.

Need to show three things to prove $W^\perp$ is a subspace.

